Example 4.10 How many licence plates may be made using either two distinct letters followed by four digits or two digits followed by 4 distinct letters where all digits or letters are distinct?

Solution:
Here we have two cases:

Case 1: The number of licence plates having two letters followed by four digits is $26 \times 25 \times 10 \times 9 \times 8 \times 7 = 32,76,000$.

Case 2: The number of licence plates having two digits followed by four letters is $10 \times 9 \times 26 \times 25 \times 24 \times 23 = 3,22,92,000$.

Since either case 1 or case 2 is possible, the total number of licence plates is $(26 \times 25 \times 10 \times 9 \times 8 \times 7) + (10 \times 9 \times 26 \times 25 \times 24 \times 23) = 3,55,68,000$.

Example 4.11 Count the number of positive integers greater than 7000 and less than 8000 which are divisible by 5, provided that no digits are repeated.

Solution:
It should be a 4-digit number greater than 7000 and less than 8000. Then the 1000th place will be the digit 7. Further, as the number must be divisible by 5 the unit place should be either 0 or 5.

7			0 or 5
1 way	8 way	7 way	2 way

Figure 4.7

As repetition is not permitted, the 100th place can be filled in 8 ways using remaining numbers and 10th place can be filled in 7 ways.

Hence, the required number of numbers is $1 \times 8 \times 7 \times 2 = 112$.

Example 4.12 How many 4 - digit even numbers can be formed using the digits $0, 1, 2, 3$ and 4, if repetition of digits are not permitted?

Solution:
There are two conditions as follows:

1. It is 4-digit number and hence its 1000th place cannot be 0.
2. It is an even number and hence its unit place can be either 0, 2 or 4.

Two cases arise in this situation. Either 0 in the unit place or not.

Case 1. When the unit place is filled by 0, then the 1000th place can be filled in 4 ways, 100th place can be filled in 3 ways and 10th place in 2 ways. Therefore, number of 4-digit numbers having 0 at unit place is $4 \times 3 \times 2 \times 1 = 24$.

			0
4 way	3 way	2 way	1 way

Figure 4.8

Case 2. When the unit place is filled with non-zero numbers, that is 2 or 4, the number of ways is 2, the number of ways of filling the 1000th place is in 3 ways (excluding '0'), 100th place in 3 ways and 10th place in 2 ways. Therefore, number of 4-digit numbers without 0 at unit place is $3 \times 3 \times 2 \times 2 = 36$.

$\overline{0}$			2/4
3 way	3 way	2 way	2 way

Figure 4.9

Hence, by the rule of sum, the required number of 4 digit even numbers is 24+36 = 60.